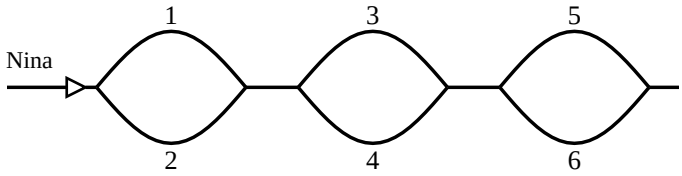
Preview: y2007_q01 (v1)

Routes through forks: one branch from every fork; number of routes = product of choices

Part A: Each question is worth 3 points

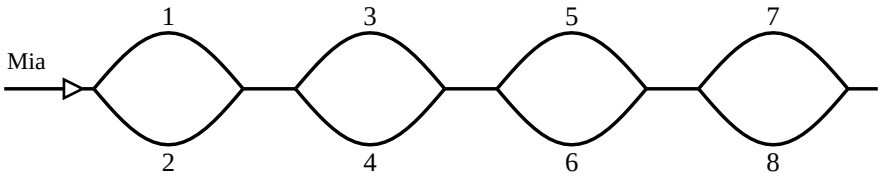
1. Nina walks along the road from left to right and writes down every number on the way.



Which group of numbers could Nina have written?

A. 1, 2 and 4 ***B.*** 1, 4 and 5 ***C.*** 1, 3 and 4 ***D.*** 4, 5 and 6 ***E.*** 3, 4 and 5

2. Mia walks along the road from left to right and writes down every number on the way.

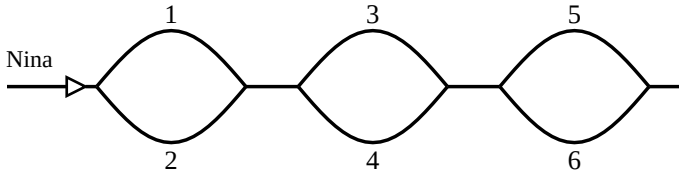


Which group of numbers could Mia have written?

A. 1, 2, 4 and 8 ***B.*** 1, 3, 4 and 5 ***C.*** 1, 4, 7 and 8

D. 1, 3, 4 and 8 ***E.*** 1, 4, 5 and 8

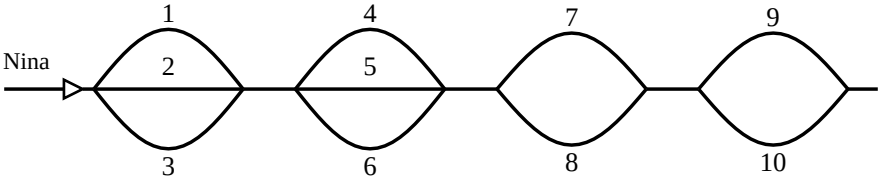
3. Nina walks along the road from left to right and writes down every number on the way.



How many different groups of numbers can Nina write?

A. 10 ***B.*** 9 ***C.*** 8 ***D.*** 6 ***E.*** 3

4. Nina walks along the road from left to right and writes down every number on the way.

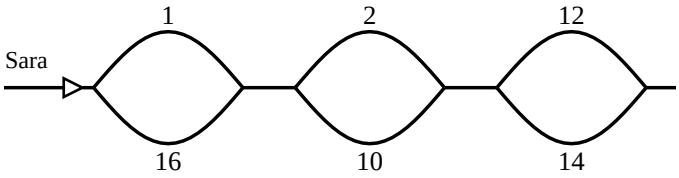


How many different groups of numbers can Nina write?

A. 45 ***B.*** 43 ***C.*** 38 ***D.*** 36 ***E.*** 18

Part B: Each question is worth 4 points

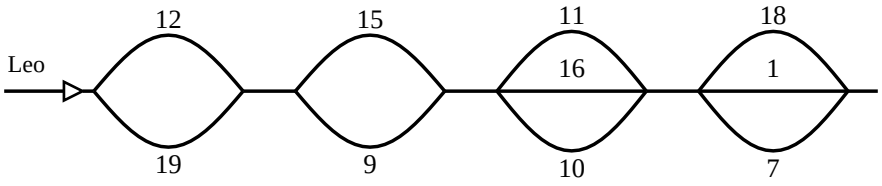
5. Sara walks along the road from left to right and adds up the numbers along the way.



What is the biggest total Sara can get?

A. 40 ***B.*** 42 ***C.*** 44 ***D.*** 49 ***E.*** 55

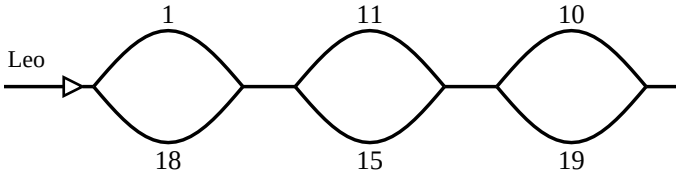
6. Leo walks along the road from left to right and adds up the numbers along the way.



What is the biggest total Leo can get?

A. 68 ***B.*** 67 ***C.*** 54 ***D.*** 52 ***E.*** 32

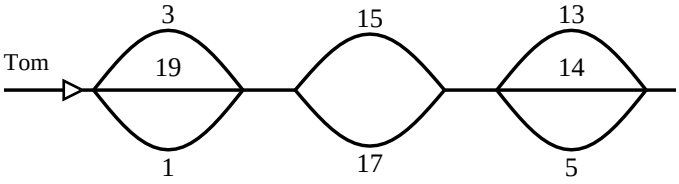
7. Leo walks along the road from left to right and adds up the numbers along the way.



Which total can Leo *not* get?

A. 22 ***B.*** 29 ***C.*** 31 ***D.*** 39 ***E.*** 48

8. Tom walks along the road from left to right and adds up the numbers along the way.

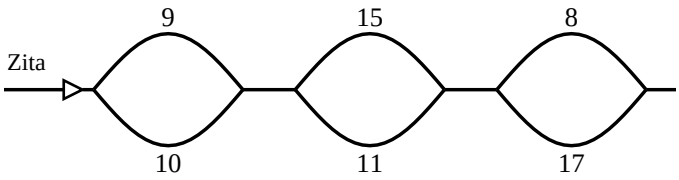


Which total can Tom *not* get?

A. 23 ***B.*** 25 ***C.*** 28 ***D.*** 30 ***E.*** 49

Part C: Each question is worth 5 points

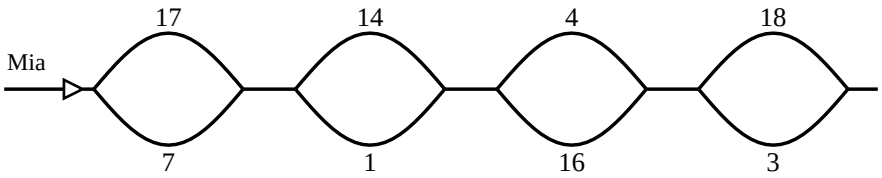
9. Zita walks along the road from left to right and adds up the numbers along the way. The total is 29.



Which number did Zita see at the second fork?

A. 8 ***B.*** 9 ***C.*** 10 ***D.*** 11 ***E.*** 15

10. Mia walks along the road from left to right and adds up the numbers along the way. The total is 28.



Which number did Mia see at the second fork?

A. 3 ***B.*** 4 ***C.*** 7 ***D.*** 14 ***E.*** 16

Answer Key

1. **B**	**2. E**	**3. C**	**4. D**	**5. A**	**6. A**
7. **B**	**8. C**	**9. D**	**10. D**		

Solutions

1. B (1, 4 and 5) — At every fork Nina takes exactly one branch, so there must be one number from each fork: 1, 4 and 5 works. Each of the others has two numbers from the same fork. [2007 Q1 · which_group]

2. E (1, 4, 5 and 8) — At every fork Mia takes exactly one branch, so there must be one number from each fork: 1, 4, 5 and 8 works. Each of the others has two numbers from the same fork. [2007 Q1 · which_group]

3. C (8) — The forks have 2, 2, 2 branches; every choice can go with every other: $2 \times 2 \times 2 = 8$. [2007 Q1 · how_many_routes]

4. D (36) — The forks have 3, 3, 2, 2 branches; every choice can go with every other: $3 \times 3 \times 2 \times 2 = 36$. [2007 Q1 · how_many_routes]

5. A (40) — Take the bigger number at every fork: 16 + 10 + 14 = 40. [2007 Q1 · largest_sum]

6. A (68) — Take the bigger number at every fork: 19 + 15 + 16 + 18 = 68. [2007 Q1 · largest_sum]

7. B (29) — 22 = 1 + 11 + 10; 31 = 1 + 11 + 19; 39 = 18 + 11 + 10; 48 = 18 + 11 + 19. No route gives 29. [2007 Q1 · impossible_total]

8. C (28) — 23 = 3 + 15 + 5; 25 = 3 + 17 + 5; 30 = 1 + 15 + 14; 49 = 19 + 17 + 13. No route gives 28. [2007 Q1 · impossible_total]

9. D (11) — Only one route gives 29: 10 + 11 + 8 = 29. At the second fork: 11. [2007 Q1 · which_branch]

10. D (14) — Only one route gives 28: 7 + 14 + 4 + 3 = 28. At the second fork: 14. [2007 Q1 · which_branch]